Standard Combinational Modules

Contemporary Digital Electronic Systems — Chapter 8

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Why This Chapter Matters

A handful of combinational patterns appear so often that they are packaged as named blocks: selecting one of several inputs, routing one input to several outputs, decoding a number into one-hot form, and finding the active line. Working with these modules is faster than gate-level design and makes schematics easier to read.

After this chapter, you can:

1. Draw and explain multiplexers, demultiplexers, decoders and encoders of common sizes.
2. Implement any n-variable Boolean function with a 2^n:1 multiplexer, or with a decoder and an OR gate.
3. Choose between a priority encoder and a basic encoder.

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How would you implement a 3-input Boolean function with no gates, using a single standard module?

What is the smallest module that always works? We answer this with multiplexers.

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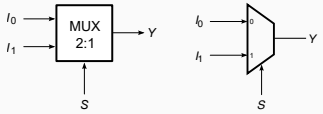
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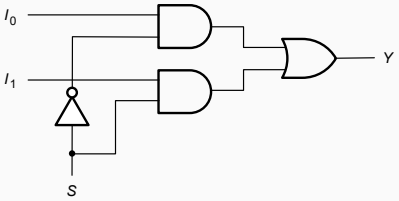
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The 2:1 Multiplexer



Rectangular and funnel symbols

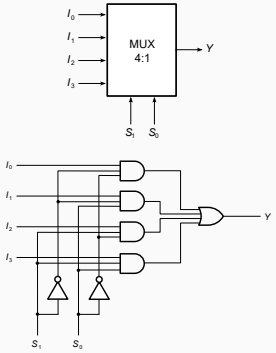


- A 2^n:1 mux has 2^n data inputs, n select lines and one output.
- $S = 0$: $Y = I_0$. $S = 1$: $Y = I_1$.

S	I_1	I_0	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$$Y = \overline{S}I_0 + SI_1$$

The 4:1 and 8:1 Multiplexers



4 AND gates, 1 OR gate, 2 inverters

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

$$Y = \overline{S_1}\,\overline{S_0} I_0 + \overline{S_1} S_0 I_1 + S_1 \overline{S_0} I_2 + S_1 S_0 I_3$$

The 8:1 mux has eight such terms, one per data input, each ANDed with its select minterm.

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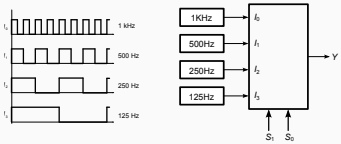
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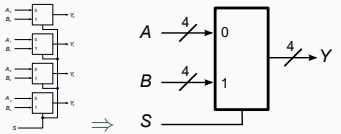
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Application: Data Selector



Choosing one of four square waves of different frequencies



- A set of four 2:1 muxes selects a whole nibble: $S = 0$ gives $Y = A$, $S = 1$ gives $Y = B$. This is a bus mux.

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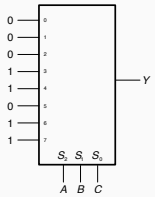
Application: Implementing Boolean Functions

- Any n-variable function can be realised with a 2^n:1 mux. No minimisation is needed.
- Connect the variables to the select lines; set each data input to the function's truth-table value.

$Y = BC + A\overline{C}$ on an 8:1 mux

Minterms m_3, m_4, m_6, m_7. Set those data inputs to 1 and the rest to 0.

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

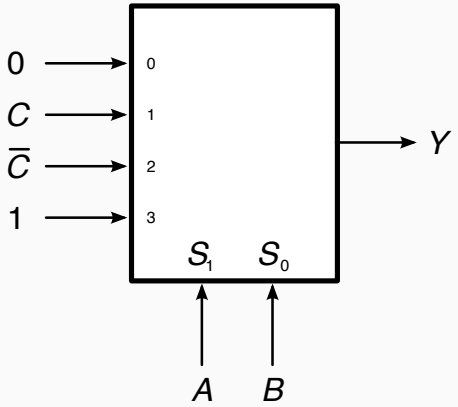


Gate-level: 4 gates. 8:1 mux: 12 gates, but no minimisation.

Using a Smaller Mux

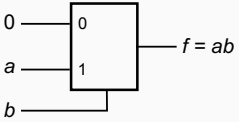
- $Y = BC + A\overline{C}$ on a 4:1 mux: A, B drive S_1, S_0.
- For each AB pair, Y is 0, 1, C or $\overline{C}$; connect that to the data input.

A	B	C	Y	Mux input
0	0	0	0	$I_0 = 0$
0	0	1	0	
0	1	0	0	$I_1 = C$
0	1	1	1	
1	0	0	1	$I_2 = \overline{C}$
1	0	1	0	
1	1	0	1	$I_3 = 1$
1	1	1	1	

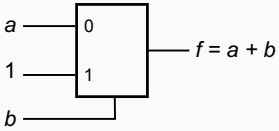


8 gates instead of 12.

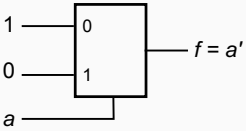
Basic Gates from a 2:1 Mux



AND: $f = \overline{b}(0) + b(a) = ab$



OR: $f = \overline{b}(a) + b(1) = a + b$



NOT: $f = \overline{a}(1) + a(0) = \overline{a}$

- Mux-based implementation suits rapid design and FPGAs, where multiplexers are core building blocks. It is not the most efficient choice for simple functions.

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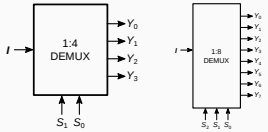
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Demultiplexers

- The opposite of a mux: one data input is routed to one of 2^n outputs chosen by n select bits.
- A mux at the transmitter and a demux at the receiver let several sender-receiver pairs share one line.



1-to-4 and 1-to-8 demultiplexers



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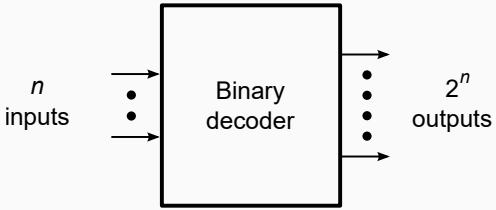
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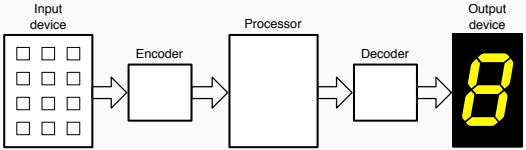
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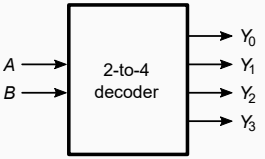
- A decoder turns an input code into a code with more bits. The common type is the n-to-2^n binary decoder: the input value activates exactly one output.
- A calculator uses encoders on input and decoders on output.

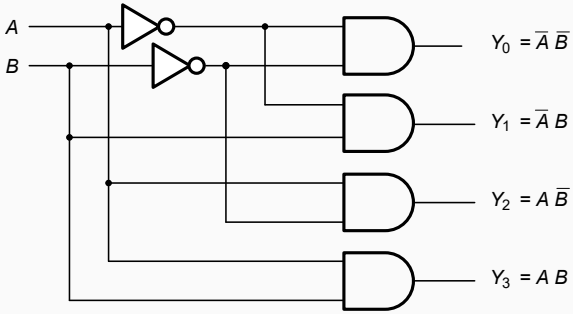




The 2-to-4 Decoder

A	B	Y_0	Y_1	Y_2	Y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1





- A 1-out-of-4 decoder produces all the minterms of two variables.
- The BCD-to-decimal decoder has 4 inputs and 10 outputs; inputs above 1001 disable all outputs.

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Decoder Applications

- Address decoding: select a device from the pattern on an address bus.
- Multiplexer control: let one of many mux inputs through.
- Demultiplexer function: send data from one input to one of many outputs.
- Implement a truth table directly: a decoder produces all 2^n minterms, so an external OR gate forms any SOP function.

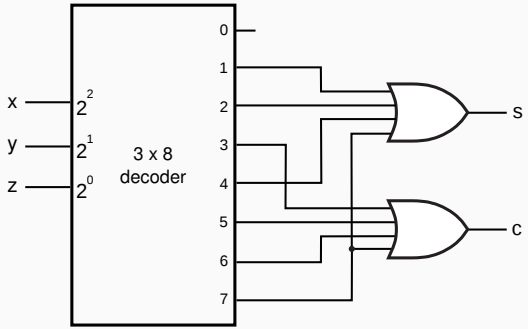
Procedure

1. Write the function as canonical SOP.
2. Pick a decoder with as many inputs as the function has variables.
3. OR together the outputs that match the function's minterms.

Example: Full Adder with a 3-to-8 Decoder

x	y	z	s	c
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$s = \sum m(1, 2, 4, 7)$$
$$c = \sum m(3, 5, 6, 7)$$



- OR decoder outputs 1, 2, 4, 7 for the sum and 3, 5, 6, 7 for the carry.
- If the decoder is built from NAND gates (active-low outputs), use NAND gates instead of OR (De Morgan).

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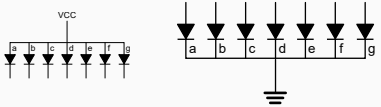
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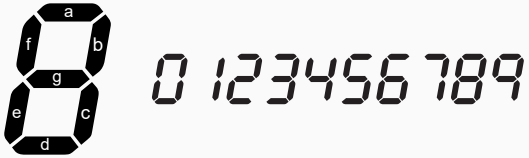
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- Seven LED segments *a* to *g* (plus an optional decimal point) show the decimal digits.
- Common anode: anodes tied together; a segment lights when its cathode is driven low.
- Common cathode: cathodes tied together; a segment lights when its anode is driven high.





Common-Anode Segment Codes

Digit	BCD	*a*	*b*	*c*	*d*	*e*	*f*	*g*
0	0000	0	0	0	0	0	0	1
1	0001	1	0	0	1	1	1	1
2	0010	0	0	1	0	0	1	0
3	0011	0	0	0	0	1	1	0
4	0100	1	0	0	1	1	0	0
5	0101	0	1	0	0	1	0	0
6	0110	0	1	0	0	0	0	0
7	0111	0	0	0	1	1	1	1
8	1000	0	0	0	0	0	0	0
9	1001	0	0	0	1	1	0	0

- Logic 0 turns a segment on; logic 1 turns it off.
- Each segment output is a function of the four BCD inputs, so a BCD-to-7-segment decoder is a combinational design problem.

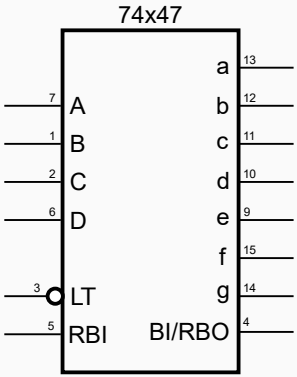
7400-Family Decoder ICs

Device	Function	Outputs	Key feature
74x138	3-to-8 decoder	Active-low	Three enable inputs
74x139	Dual 2-to-4 decoder	Active-low	Two independent channels
74x42	BCD-to-decimal	Active-low	10 mutually exclusive outputs
74x47	BCD-to-7-segment	Active-low	Open-collector, common anode
74x48	BCD-to-7-segment	Active-high	Internal pull-ups, common cathode

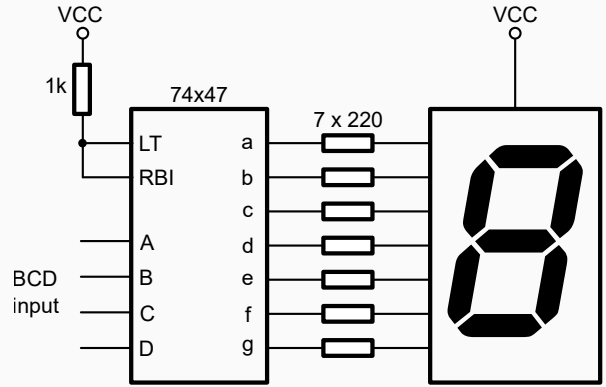
- Programmable logic and microcontrollers are alternatives, but these ICs remain useful for learning, driving displays and maintaining existing circuits.
- Check the datasheet: output circuitry and electrical characteristics differ.

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The 74x47 Driver



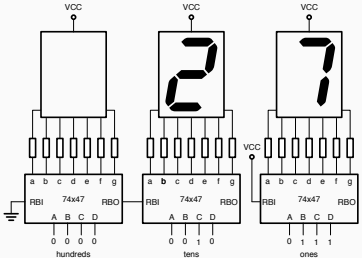
LT	Lamp test
RBI	Ripple blanking in
RBO	Ripple blanking out



- For one digit: pull LT and RBI high. Never leave control inputs floating.
- Use a current-limiting resistor per segment, unless the display module has them.

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Leading-Zero Suppression



- When RBI is low and the BCD input is 0000, the decoder blanks the display and drives RBO low.
- Connect RBO of the hundreds digit to RBI of the tens digit, and RBO of the tens digit to RBI of the ones digit.
- Hold the hundreds RBI low (suppress) and the ones RBI high, so at least one digit always shows.

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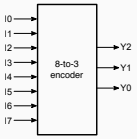
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- The opposite of a decoder: 2^n input lines are compressed into an n-bit code.
- An 8-to-3 encoder outputs the binary index of the single active input.
- In an ideal encoder, only one input may be active at a time.

I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	Y_2	Y_1	Y_0
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

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Priority Encoders

- If several inputs can be active together, a plain encoder gives undefined results.
- A priority encoder considers only the highest-priority active input.
- Used where many devices may ask for attention at once but only one is served at a time.

I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	Y_2	Y_1	Y_0
1	0	0	0	0	0	0	0	0	0	0
×	1	0	0	0	0	0	0	0	0	1
×	×	1	0	0	0	0	0	0	1	0
×	×	×	1	0	0	0	0	0	1	1
×	×	×	×	1	0	0	0	1	0	0
×	×	×	×	×	1	0	0	1	0	1
×	×	×	×	×	×	1	0	1	1	0
×	×	×	×	×	×	×	1	1	1	1

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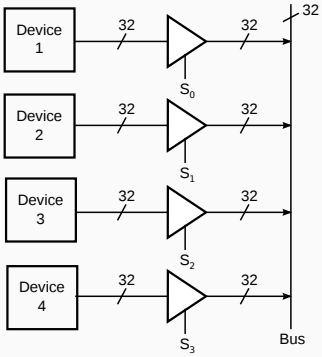
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Tri-State Buffers as a Mux

- Each tri-state buffer has an enable: on, it drives the bus; off, it is high impedance (disconnected).
- With exactly one buffer enabled, the bus carries one source: a mux at the wire level.
- A decoder makes the enables, so at most one buffer drives the bus. This prevents contention and short circuits.



Back to the Opening Question

A 3-input function with no gates

An 8:1 multiplexer. Wire the three inputs to the select lines and hard-wire each data input to the truth-table value (0 or 1). Any 3-variable function works, so 8:1 is the smallest mux that *always* works.

With a little more care (using C or $\overline{C}$ on the data inputs) a 4:1 mux can implement many of them too.

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Summary

- A 2^n:1 mux selects one of 2^n inputs with n select bits; a demux routes one input to one of 2^n outputs.
- A decoder asserts exactly one of 2^n outputs: a one-hot translator.
- A 2^n:1 mux can implement any n-variable function by wiring its data inputs to the truth-table values.
- A priority encoder returns the index of the highest-priority active input; a plain encoder needs at most one active input.
- Tri-state buffers on a shared bus form a multi-source mux; only one driver may be enabled at a time.

Looking ahead

Chapter 9 shows how arithmetic circuits combine bits to calculate sums, differences and comparisons.